PROJECT SPARK — LEVEL 12 CREATION PROCESS

Series–Parallel Circuit Challenge | Unity 6 | Build checklist

Goal: Make a new Level 12 prefab and complete four objectives using the already-working universal workstation binding architecture. **Do not replace any binding, solver, shelf, or evaluator C# scripts.**

0 — Back up and prepare

- Back up the completed Level 11 prefab and **Level_11.asset**.
- Exit Play Mode. Build and configure the **original** **Level_12_Layout** prefab, not a **(Clone)**.
- Confirm **Level_12.asset** exists at **Assets/My_Assets/_Project_Spark/Prefabs/Levels/Level_12_Layout/Level_12.asset**.
- Keep the Level 11 prefab unchanged and reuse only its successful setup patterns.

1 — Import Level 12 mission and objectives

- In Project, select the **original** **Level_12.asset**.
- Use your existing Project Spark editable-ScriptableObject TXT importer to import **Level_12_SeriesParallel_Editable_Import.txt** (included).
- Verify Level ID **level_12**, title **Series–Parallel Circuit Challenge**, and target count **4**.
- Check the four target IDs: **series_parallel_r1_conducting**, **series_parallel_r2_current**, **series_parallel_r3_voltage**, **series_parallel_source_current**.
- Save the asset. Importing this TXT does **not** create physical GameObjects, wires, or binding references.

2 — Build the original Level_12_Layout prefab

- Open **Level_12_Layout** in Prefab Mode. Create or adapt the **layout**, without changing Level 11.
- On the prefab root: **SparkLevelLayout**, **one primary** **SparkLevelSceneBindings**, and **one** **SparkLevelWorkstationBindingBridge**.
- Set **SparkLevelLayout.Level Definition = Level_12** and **Scene Bindings** to its own primary bindings (same level prefab).
- Add/keep a 5 V **PowerSupply_01**, **Power_Switch**, LCD measurement terminals, fixed **Resistor_330_Ω** (R1), fixed **Resistor_470_Ω** (R3), and a **workstation** **SparkShelfSlot** for replaceable R2.
- Set actual fixed resistor resistances to **330** Ω and **470** Ω. Level 11 taught us not to trust labels alone.
- Ensure the R2 workstation **SparkShelfSlot.Kind** is **Workstation**, its category permits resistors, and **SlotId** is unique among active workstation slots. **Register_Slot_2** is only an example; use the real Level 12 slot's ID.
- Keep the 220 Ω / 100 Ω shelf items in the **persistent** **Component_Shelf** rather than copying them into the Level 12 prefab.

3 — Wire the intended topology

```
Battery +5V --- Switch ----+---- [R1 330Ω] -------------------+
                           |                                  |
                           +---- [R2 220Ω] ---- [R3 470Ω] ----+
                                                              |
Battery negative ---------------------------------------------+

LCD Terminal A -> R3 input (junction of R2 output and R3 input)
LCD Terminal B -> R3 output (battery negative return)
```

Connection checklist (eight actions):

- Battery positive -> Switch input.
- Switch output -> R1 (330 Ω) input.
- Switch output -> R2 workstation resistor input.
- R2 output -> R3 (470 Ω) input (**series link**, critical).
- R1 output -> Battery negative.
- R3 output -> Battery negative.
- LCD Terminal A -> R3 input.
- LCD Terminal B -> R3 output.

Never connect R2 output directly to battery negative; that would incorrectly make R2 parallel rather than series with R3. The LCD must measure, not bypass, R3.

4 — Author the fixed binding IDs (same level prefab)

Use the primary **SparkLevelSceneBindings** on **Level_12_Layout**. These exact IDs are referenced by the included TXT:

Category	ID	Scene object
Terminal	**Battery_positive**	Supply positive terminal
Terminal	**Battery_negative**	Supply negative terminal
Terminal	**Resistor_330_Input**	R1 input terminal
Terminal	**Resistor_330_Output**	R1 output terminal
Terminal	**Resistor_470_Input**	R3 input terminal
Terminal	**Resistor_470_Output**	R3 output terminal
Component	**PowerSupply_01**	5 V SparkPowerSupply
Component	**Resistor_330_Ω**	330 Ω SparkResistor
Component	**Resistor_470_Ω**	470 Ω SparkResistor

- Reassign these fixed references to objects **inside Level 12**; copied Level 11 references must not point at its old prefab.
- Keep permanent battery, switch, LCD bindings as needed by your existing systems.
- Do **not** add **R2_Component**, **R2_Input**, or **R2_Output** as fixed references; they are runtime declarations.

5 — Configure the Universal Workstation Binding Bridge

Attach the bridge to the **original Level_12_Layout root**, not to **Component_Shelf**. Configure Workstations > Element 0:

Inspector field	Enter
Slot Id	Exact **SparkShelfSlot.SlotId** of the Level 12 R2 workstation
Component Binding Id	**R2_Component**
Input Terminal Binding Id	**R2_Input**
Output Terminal Binding Id	**R2_Output**
Electronic Object Binding Id	leave blank

Inspector field	Enter
Electrical Type Name	**SparkResistor**
Electronic Type Name	leave blank
Input Terminal Name	**_Input**
Output Terminal Name	**_Output**
Exact Terminal Names	unchecked
Resync Seconds	**0.25**

- In **Edit Mode** (Prefab Mode), click **DECLARE / CHECK RUNTIME IDS IN PRIMARY BINDINGS**.
- Verify **runtimeComponentIds** contains **R2_Component**, and **runtimeTerminalIds** contains **R2_Input**, **R2_Output**.
- Save the original prefab. Do not rely on changes to a runtime **(Clone)** to persist.
- Leave the shelf where it is. The bridge uses the workstation slot's ID, **not** the inventory/storage position (**Placement_01_04** is storage).

6 — Level manager and loader integration

- Ensure the layout loader's Level 12 entry maps **Level_12** to the **Level_12_Layout prefab**.
- Ensure LevelGamePlayManager's **levels** array includes **Level_12** at the intended index.
- Ensure progression/unlocking allows entering Level 12 after completing Level 11.
- Ensure the shelf registers the level's workstation slot on layout load, using the existing **RegisterRuntimeSlots** flow.
- Never add a second bridge writing the same R2 runtime IDs.

7 — Player journey / initial state

- Start with R2 uninstalled or with **100** Ω installed as the intentionally incorrect value.
- Require player to switch supply OFF before replacement (according to your existing shelf interaction policy).
- Remove 100 Ω and insert the real shelf **220** Ω resistor into the Level 12 workstation.
- Attach the series link between R2 output and R3 input, then complete return and LCD probe connections.
- Switch on and check objectives.

8 — Expected measurements at 5.000 V

Quantity	With correct R2 = 220 Ω	With wrong R2 = 100 Ω
R1 current (330 Ω)	15.152 mA	15.152 mA
R2/R3 series-branch current	7.246 mA	8.772 mA
R3 voltage (470 Ω)	3.406 V	4.123 V
Total source current	22.398 mA	23.923 mA

The intended 220 Ω circuit satisfies the TXT's R2 current (7.0–7.5 mA), R3 voltage (3.30–3.50 V), and source-current (21.5–23.0 mA) targets. These are ideal calculations; actual solved values can differ slightly due to modeled switch/wire resistance.

9 — Play Mode acceptance checklist

- Level 12 loads as **Level_12_Layout(Clone)** and no binding validation error is printed.
- Workstation bridge log reports **slot=<LEVEL_12_WORKSTATION_ID> | item=Resistor_220_Ω | source=SLOT_OCCUPANT | bound=Resistor_220_Ω** (names may vary).
- **R2_Component**, **R2_Input**, **R2_Output** resolve to the actual installed 220 Ω shelf item.
- R1 current about 15.15 mA, R2 current about 7.25 mA, R3 voltage about 3.41 V, source current about 22.40 mA.
- No source short, target short, or overload; closed return = true.
- All four targets pass and Level 12 completes.
- Exit Play Mode; original prefab IDs remain saved.

10 — Fault isolation (without modifying working C#)

Symptom	Check first
SLOT_MISSING	Slot ID and **Kind=Workstation**; registered in **SparkObjectShelf**
item=EMPTY	Is resistor actually installed in the workstation, not on storage shelf?
R2_* UNBOUND	Primary runtime declarations, suffixes **_Input** / **_Output**, single bridge
InvalidConfiguration	Fixed Level 12 references/IDs in primary SceneBindings, including **PowerSupply_01**
R2 current ~22.7 mA	R2 was accidentally placed directly across 5 V, not in series with R3
R3 voltage ~5 V	R3 may be connected directly across supply, not after R2
Source current ~48.5 mA	Level 11 parallel topology was accidentally retained
3/4 targets	Inspect target result logs to identify the exact failing objective
LCD unexpectedly alters current	Check that LCD is wired as a measurement across R3, not a conducting bypass

Scope note: This package is a build procedure plus the Level 12 asset TXT. It does not automatically construct Unity GameObjects or place wires. Existing Unity editor and runtime systems remain untouched.